

Lecture 2:

Mathematical Analysis

1 Intervals

Definition 1.1 (Interval). [def:interval] A set $I \subseteq \mathbb{R}$ is an *interval* if, whenever $x, y \in I$ with $x \leq y$, every $z \in \mathbb{R}$ satisfying $x \leq z \leq y$ also belongs to I .

Remark 1.1 (Intervals as Sets). [rem:intervals-as-sets] In these notes, every interval is a set contained in $\mathbb{R}$; that is, every interval I satisfies $I \subseteq \mathbb{R}$.

Definition 1.2 (Open Interval). [def:open-interval] For $a, b \in \mathbb{R}$ with $a < b$, the *open interval* from a to b is

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Definition 1.3 (Closed Interval). [def:closed-interval] For $a, b \in \mathbb{R}$ with $a \leq b$, the *closed interval* from a to b is

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Lemma 1.1 (Endpoints of Contained Closed Intervals). [lem:contained-interval-endpoints] Let $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ be closed intervals. If $I_1 \subseteq I_2$, then

$$a_2 \leq a_1 \leq b_1 \leq b_2.$$

Proof. Since $a_1, b_1 \in I_1 \subseteq I_2$, definition 1.3 gives $a_2 \leq a_1$ and $b_1 \leq b_2$. Since $I_1 = [a_1, b_1]$ is a closed interval, definition 1.3 requires $a_1 \leq b_1$. Combining these inequalities gives the result. □

2 Nested Intervals Property

Theorem 2.1 (Nested Intervals Property (NIP)). [thm:nip] Let $I_n = [a_n, b_n]$ be a sequence of closed and bounded intervals.

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots \implies \bigcap_{n=0}^{\infty} I_n \neq \emptyset.$$

Proof. Take $A = \{a_n : n \in \mathbb{N}\}$ and $B = \{b_n : n \in \mathbb{N}\}$. Since I_n is a closed interval, we have $a_n \leq b_n$ and

$$I_n = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}.$$

Also,

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots, \quad \text{that is,} \quad I_{n+1} \subseteq I_n$$

for every $n \in \mathbb{N}$.

For a fixed $k \in \mathbb{N}$, suppose

$$x \in \bigcap_{n=0}^k I_n.$$

By the definition of intersection, $x \in I_n$ for every $0 \leq n \leq k$. In particular, taking $n = k$ gives

$$x \in I_k = [a_k, b_k],$$

and hence $a_k \leq x \leq b_k$.

Since A is a set in $\mathbb{R}$, according to Axiom of Completeness, A has a supremum, say $\sup A = \alpha$. □